AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MT101
General Mathematics

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 7 /April /2025

Number of Exam Pages:2 (including this cover sheet)	Time Allowed: 90 minutes

Instructions:

- Total Marks:**60 marks**
- This exam consists of **4 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Scientific Calculators** are allowed.

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ESSAY QUESTIONS

Each question is worth 15 marks. You should answer all the questions, giving 60 possible marks.

Q–1: [4+5+6 Marks]

a) let f(x)= x-9, and h(x)= $\sqrt{x}$

i) Evaluate (f + h) (-4)

ii) Evaluate (f o h) (x). Write the domain of (f o g) (x).

b) Solve for x: 6(x—1) +4 = 7x+1

Q–2: [8+7Marks] solve the following:

a) 3 | x+3 |- 21 >0 (write solution in interval notation)

b) $-10 \leq 4 - 3x < 2$ (write solution in interval notation)

Q–3: [8+7] Marks] a) solve the following equations:

a) $\sqrt{2x + 7} - x = 2$

b) x^2 -2x+ 2 =0 (use the quadratic formula)

Q–4: [8+7Marks]

a) Write the equation of a circle in standard form, and identify the center and the radius. $2x^2 + 2y^2 - 16y+6=0$

b) Find the equation of the line that passes through the two points A(0, -1),and B(5, 9).

End of the questions.

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